

SOK-1301 H25 Forelesning 3

tidyverse og ggplot2

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1. Innledning

- **Mål:** Bruke grunnleggende kommandoer fra dplyr og ggplot2 til å lage figurer
- **Transformere data:** Fem kommandoer i dplyr: `select()`, `filter()`, `arrange()`, `mutate()`, `summarize()`
- **Visualisere data:** Vi bruker ggplot2 i kombinasjon med dplyr
- **Gjennomføring:** Jeg introduserer kommandoene, dere prøver selv ved hjelp av vedlagt kode

2. Koden til forelesning 3 SOK-1301_Forelesning_3.R

2.1 Lagre data i en tibble

En tibble er det anbefalte lagringsformatet i tidyverse. Datasettet kommer i en .csv-fil, og tidyverse lagrer dette automatisk som tibble (tbl). Skriv `class(co2data)` for å sjekke.

```
# rydd opp
rm(list=ls())

# last inn tidyverse
# sikre at oppstart på pakken ikke vises i pdf
suppressPackageStartupMessages(library(tidyverse))

#####
### data i tibble-format ###
#####
```

```
# les CO2 data i .csv fra OWID
url <- "https://raw.githubusercontent.com/owid/co2-data/master/owid-co2-data.csv"
co2data <- read_csv(url)
```

```
Rows: 50191 Columns: 79
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (2): country, iso_code
```

```
dbl (77): year, population, gdp, cement_co2, cement_co2_per_capita, co2, co2...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# se: https://github.com/owid/co2-data
# se en beskrivelse av data her:
# https://github.com/owid/co2-data/blob/master/owid-co2-codebook.csv

# se på dataene i konsollen
co2data
```

```
# A tibble: 50,191 x 79
```

	country	year	iso_code	population	gdp	cement_co2	cement_co2_per_capita
	<chr>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	Afghanistan	1750	AFG	2802560	NA	0	0
2	Afghanistan	1751	AFG	NA	NA	0	NA
3	Afghanistan	1752	AFG	NA	NA	0	NA
4	Afghanistan	1753	AFG	NA	NA	0	NA
5	Afghanistan	1754	AFG	NA	NA	0	NA
6	Afghanistan	1755	AFG	NA	NA	0	NA
7	Afghanistan	1756	AFG	NA	NA	0	NA
8	Afghanistan	1757	AFG	NA	NA	0	NA
9	Afghanistan	1758	AFG	NA	NA	0	NA
10	Afghanistan	1759	AFG	NA	NA	0	NA

```
# i 50,181 more rows
```

```
# i 72 more variables: co2 <dbl>, co2_growth_abs <dbl>, co2_growth_prct <dbl>,
# co2_including_luc <dbl>, co2_including_luc_growth_abs <dbl>,
# co2_including_luc_growth_prct <dbl>, co2_including_luc_per_capita <dbl>,
# co2_including_luc_per_gdp <dbl>, co2_including_luc_per_unit_energy <dbl>,
# co2_per_capita <dbl>, co2_per_gdp <dbl>, co2_per_unit_energy <dbl>,
# coal_co2 <dbl>, coal_co2_per_capita <dbl>, consumption_co2 <dbl>, ...
```

2.2 Velg variabel med `select()`

Bruk `select()` til å velge variable (kolonner) som egen tabell:

```
select(co2data, iso_code)
```

```
# A tibble: 50,191 x 1
  iso_code
  <chr>
1 AFG
2 AFG
3 AFG
4 AFG
5 AFG
6 AFG
7 AFG
8 AFG
9 AFG
10 AFG
# i 50,181 more rows
```

2.3 Velg verdier med `filter()`

Bruk `filter()` til å velge verdier (rader) som egen tabell:

```
filter(co2data, iso_code == "AFG")
```

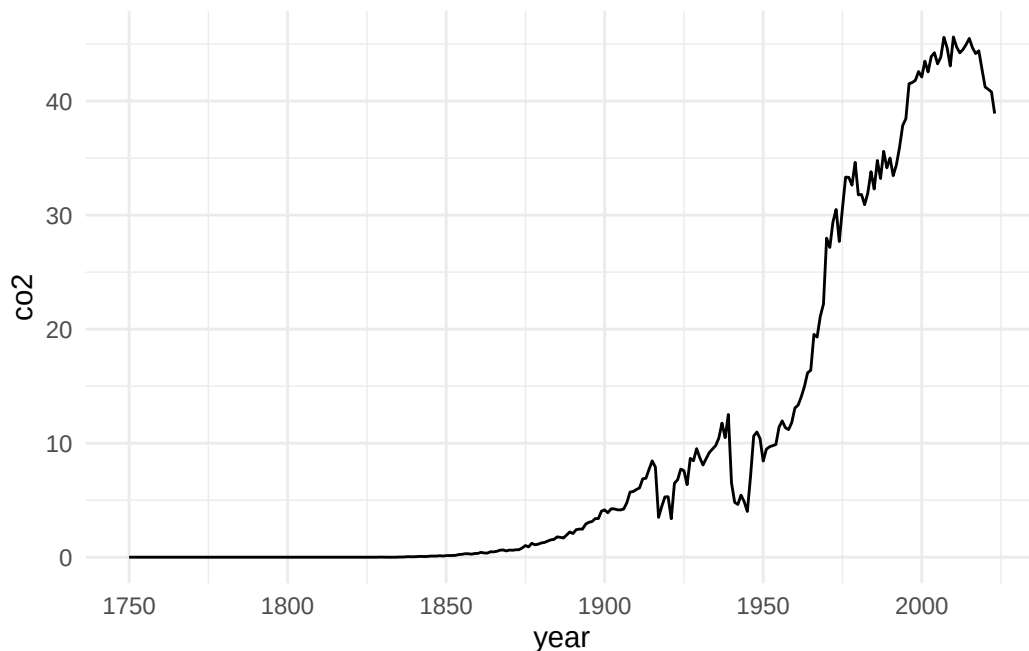
```
# A tibble: 274 x 79
  country      year iso_code population    gdp cement_co2 cement_co2_per_capita
  <chr>      <dbl> <chr>      <dbl> <dbl>      <dbl>              <dbl>
1 Afghanistan 1750 AFG          2802560    NA          0              0
2 Afghanistan 1751 AFG              NA    NA          0              NA
3 Afghanistan 1752 AFG              NA    NA          0              NA
4 Afghanistan 1753 AFG              NA    NA          0              NA
5 Afghanistan 1754 AFG              NA    NA          0              NA
6 Afghanistan 1755 AFG              NA    NA          0              NA
7 Afghanistan 1756 AFG              NA    NA          0              NA
8 Afghanistan 1757 AFG              NA    NA          0              NA
9 Afghanistan 1758 AFG              NA    NA          0              NA
10 Afghanistan 1759 AFG              NA    NA          0              NA
# i 264 more rows
```

```
# i 72 more variables: co2 <dbl>, co2_growth_abs <dbl>, co2_growth_prct <dbl>,
#   co2_including_luc <dbl>, co2_including_luc_growth_abs <dbl>,
#   co2_including_luc_growth_prct <dbl>, co2_including_luc_per_capita <dbl>,
#   co2_including_luc_per_gdp <dbl>, co2_including_luc_per_unit_energy <dbl>,
#   co2_per_capita <dbl>, co2_per_gdp <dbl>, co2_per_unit_energy <dbl>,
#   coal_co2 <dbl>, coal_co2_per_capita <dbl>, consumption_co2 <dbl>, ...
```

2.4 Bruk ggplot2 til å lage figurer

- Data - aes() (koordinatsystem) - og geom (visualisering, farger, linjestørrelse) =)
- Råd: Bruk cheat sheet og pipes

```
co2data %>%
  filter(country == "Norway") %>%
  ggplot(aes(x=year, y=co2)) %>%
  + geom_line() %>%
  + theme_minimal()
```



2.5 Gjør oppgave 1-4 i koden til forelesning 3

2.6 Endre rekkefølge på verdier med arrange()

Bruk arrange() til å endre rekkefølgen på verdier:

```
arrange(co2data, co2)
```

```
# A tibble: 50,191 x 79
  country      year iso_code population  gdp cement_co2 cement_co2_per_capita
  <chr>      <dbl> <chr>      <dbl> <dbl>      <dbl>          <dbl>
1 Africa (GCP) 1850 <NA>      NA     NA         NA             NA
2 Africa (GCP) 1851 <NA>      NA     NA         NA             NA
3 Africa (GCP) 1852 <NA>      NA     NA         NA             NA
4 Africa (GCP) 1853 <NA>      NA     NA         NA             NA
5 Africa (GCP) 1854 <NA>      NA     NA         NA             NA
6 Africa (GCP) 1855 <NA>      NA     NA         NA             NA
7 Africa (GCP) 1856 <NA>      NA     NA         NA             NA
8 Africa (GCP) 1857 <NA>      NA     NA         NA             NA
9 Africa (GCP) 1858 <NA>      NA     NA         NA             NA
10 Africa (GCP) 1859 <NA>      NA     NA         NA             NA
# i 50,181 more rows
# i 72 more variables: co2 <dbl>, co2_growth_abs <dbl>, co2_growth_prct <dbl>,
#   co2_including_luc <dbl>, co2_including_luc_growth_abs <dbl>,
#   co2_including_luc_growth_prct <dbl>, co2_including_luc_per_capita <dbl>,
#   co2_including_luc_per_gdp <dbl>, co2_including_luc_per_unit_energy <dbl>,
#   co2_per_capita <dbl>, co2_per_gdp <dbl>, co2_per_unit_energy <dbl>,
#   coal_co2 <dbl>, coal_co2_per_capita <dbl>, consumption_co2 <dbl>, ...
```

2.7 Lag nye variable med mutate()

```
mutate(co2data, x= year + 1)
```

```
# A tibble: 50,191 x 80
  country      year iso_code population  gdp cement_co2 cement_co2_per_capita
  <chr>      <dbl> <chr>      <dbl> <dbl>      <dbl>          <dbl>
1 Afghanistan 1750 AFG      2802560  NA         0             0
```

```

2 Afghanistan 1751 AFG      NA      NA      0      NA
3 Afghanistan 1752 AFG      NA      NA      0      NA
4 Afghanistan 1753 AFG      NA      NA      0      NA
5 Afghanistan 1754 AFG      NA      NA      0      NA
6 Afghanistan 1755 AFG      NA      NA      0      NA
7 Afghanistan 1756 AFG      NA      NA      0      NA
8 Afghanistan 1757 AFG      NA      NA      0      NA
9 Afghanistan 1758 AFG      NA      NA      0      NA
10 Afghanistan 1759 AFG      NA      NA      0      NA
# i 50,181 more rows
# i 73 more variables: co2 <dbl>, co2_growth_abs <dbl>, co2_growth_prct <dbl>,
#   co2_including_luc <dbl>, co2_including_luc_growth_abs <dbl>,
#   co2_including_luc_growth_prct <dbl>, co2_including_luc_per_capita <dbl>,
#   co2_including_luc_per_gdp <dbl>, co2_including_luc_per_unit_energy <dbl>,
#   co2_per_capita <dbl>, co2_per_gdp <dbl>, co2_per_unit_energy <dbl>,
#   coal_co2 <dbl>, coal_co2_per_capita <dbl>, consumption_co2 <dbl>, ...

```

2.8 Få deskriptiv statistikk med summarize()

Bruk `summarize()` til å få en tabell med deskriptiv statistikk, eksempelvis sum, gjennomsnitt:

```
summarize(group_by(co2data, year), tot_co2 = sum(co2))
```

```

# A tibble: 274 x 2
   year tot_co2
   <dbl>   <dbl>
1  1750     NA
2  1751     NA
3  1752     NA
4  1753     NA
5  1754     NA
6  1755     NA
7  1756     NA
8  1757     NA
9  1758     NA
10 1759     NA
# i 264 more rows

```

2.9 Gjør oppgave 5-8 i koden fra forelesning 3